w.e.f. 2022-23

Program: Program: B. Tech Computer Science and Engineering (Cybersecurity)	**Semester : VI**
Course : Application Security Testing	**Code :**

Teaching Scheme				**Evaluation Scheme**	
Lecture (Hours per week)	**Practical (Hours per week)**	**Tutorial (Hours per week)**	**Credit**	**Internal Continuous Assessment (ICA) (Marks- 50)**	**Term End Examinations (TEE) (Marks- 100)**
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50

Prerequisite: Web Programming, Cybersecurity,

Course Objective: This course introduces students to an exciting field of security testing. Various security testing frameworks and methods will be discussed in the course. Students will perform security testing of web and mobile application and suggest countermeasures.

Course Outcomes

After completion of the course, the student will be able to -

1. Describe various testing methodologies
2. Perform security testing of web and mobile applications
3. Suggest countermeasure to mitigate vulnerabilities

Detailed Syllabus

Unit	Description	Duration
1.	**Introduction:** Application architecture (MVC, Micro-services, Serverless, SPAs), Types of application security testing (SAST, DAST, IAST) (16.2) [https://www.imperva.com/learn/application-security/application-security/]	04
2.	**Secure Software Lifecycle:** Need (16.1), SecSDL (Microsoft SDL, Touchpoints , SAFECode (16.2), adaption of SecSDL to agile, mobile and cloud environment(16.3.1-16.3.3), Assessing the Secure software lifecycle (SAMM, BSIMM, Common Criteria) (16.4).	05
3.	**Software Security:** Software vulnerabilities and its types and countermeasures, software vulnerabilities detection mechanism, Testing (14) Frameworks (NIST, OWASP[https://www.imperva.com/learn/application-security/owasp-top-10/], CWE), CVSS, OWASP top 10 web Vulnerabilities(ch.2)	05
4.	**Web App Security and Testing:** Pentesting Methodology (ch.3), testing of authentication(ch.4), session management(ch.5), secure channels (ch.6), secure access control(ch.7), data and information disclosure(ch.8), injection attacks, CSRF [https://www.imperva.com/learn/application-security/csrf-cross-site-request-forgery/], Server side template injection, clickjacking[https://www.imperva.com/learn/application-security/clickjacking/], Configuration and Deployment, API testing, appropriate countermeasures to mitigate vulnerabilities.	10
5.	**Mobile App Security and Testing:** Types of mobile apps, challenges in mobile testing, OWASP mobile top 10 testing, appropriate countermeasures to mitigate vulnerabilities.	06
	Total	**30**

Text Books

1. **Richa Gupta *Hands-on Penetration Testing for Web Applications*, 1st Edition, BPB Publications, 2021.**

Reference Books

1. Tanya Janca, *Alice & Bob Learn Application Security*, Wiley, 2021.
2. Rashid et al., *The Cyber Security Body of Knowledge, CyBOK, v1.0*, 2019. Available Online: **https://www.cybok.org/media/downloads/CyBOK-version-1.0.pdf**
3. Andrew Hoffman, *Web Application Security*, O'Reilly Media, 2020.
4. Vijay Kumar Velu, *Mobile Application Penetration Testing,* Packt Publications, 2016

Laboratory Work

8 to 10 lab exercises (and a practicum) based on the syllabus.

Signature

(Head of the Department)

Signature

(Dean)